

Q1. $E(x_i) = 1(0.5) + 0(0.5) = 0.5$

(a) $\sigma^2 = E(x^2) - E(x)^2 = 0.5 - (0.5)^2 = 0.25$

Since $S_n = X_1 + X_2 + \dots + X_n$

$E(S_n) = \sum E(x_i) = n \cdot 0.5 = n/2$

$E[\text{var}(S_n)] = \sum \text{Var}(x_i) = n \cdot 0.25 = n/4$

(b) $Z_n = \frac{S_n - E(S_n)}{\sqrt{\text{var}(S_n)}} = \frac{S_n - n/2}{\sqrt{n/4}}$

(f) as $n \rightarrow \infty$, shape of histogram smoothens, symmetric & bell shaped. For $n=10$, there are "steps" because S_n take int values. As $n \rightarrow 50$, clustering & densely together & shrinking the gaps

(h) The exp visually and illustrates CLT. Resulting variables Z_n clearly adapted the shape of standard normal distribution $N(0,1)$.

Q2
(a)

$$E(g(y)) = \int_{-\infty}^{\infty} g(y)f(y)dy.$$

$$E(e^{-y^2}) = \int_0^1 e^{-y^2}(1)dy = \int_0^1 e^{-x^2}dx = I.$$

Integral is exactly eqv. to expected value of e^{-y^2} .

(e) the approx. improves due to Law of Large No., which guarantees that as $N \rightarrow \infty$, sample $\bar{x}$ converges to true exp. value.

Further, by CLT, error of the Monte Carlo estimate is $\propto \frac{1}{\sqrt{N}}$. As $N \uparrow$ error $\downarrow$.

(f) If we use a grid with m points per dimension to eval. an integral in d dimension, we evaluate funcⁿ m^d times. The comp cost grows exp.

Monte Carlo integration avoids this. Because error of a Monte Carlo int. depends upon the N and variance of funcⁿ, error is independent of dimension d .

Q3
(a)

$$\text{Total area} = 2 \times 2 = 4.$$

$$\text{Req. area} = \pi \times 1^2 = \pi.$$

$$P = \frac{\pi}{4}$$

Q36) Due to law of large numbers. As N grows, the empirical proportion of successes converge towards the true prob. p .

Q4
(9) $S_T = S_0 e^{\sigma Z}$, because S_T is function of z ($\because S_T = f(z)$) it is also stochastic.

(b) Volatility affects simulation by following:-

- Speed of terminal stock prices, distribution of S_T becomes much wider and features fatter tails.
- As volatility $\uparrow$, expected option payoff $\uparrow$

- (c) (1) flexibility with complex payoffs
(2) High dimensionality
(3) Advanced stochastic models